Course: MAT 112 – Linear Algebra and Differential Equations

Department of Software Engineering, Metropolitan University

5 questions will be given, 4 should be answered

Topics –

1. Complex Variables: (1 set)
    - Expressing into polar form. (Don't forget to check the position of a point in the quadrant, to determine the angle θ properly.)
    - Roots of complex numbers: Example: 1.28, 1.29, 1.30, 1.37
2. Differential Equation:
    A. Ordinary Differential Equation (2 sets from this part)
        - Determining different types of D.E and Solution of those differential equations – (i) Variable Separable Equations (ii) Homogeneous ODE (iii) Linear ODE (iii) Exact ODE (all of first order.)
          ---Practice the problems solved in the class and given as homework---
          From Book:
          <u>Variable Separable and Homogenous Equation:</u> Schaum's Outlines: Page 22 – 4.1-4.3, 4.5-4.7, 4.10)
          <u>Linear ODE and Exact ODE:</u> BD Sharma --- Page 18 of Part I: Ex-1, 2,3,10
          Page 37 of part I: Ex – 1, 3, 6
        - Linear differential equations with constant coefficients: (of Higher Order)
          <u>From BD Sharma:</u> Page 63 of Part I – 1,2,3.
3. Matrix (2 Set)

Determining Inverse –

(1) If $A = \begin{bmatrix} -1 & 2 & -3 \\ 2 & 1 & 0 \\ 4 & -2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix}$ find $A^{-1}B$.

(2) Find the inverse of matrix $A = \begin{bmatrix} 5 & -6 & 4 \\ 7 & 4 & -3 \\ 2 & 1 & 6 \end{bmatrix}$.

(3) Find the inverse of the matrix $A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -1 \\ 3 & 3 & 2 \end{bmatrix}$.

Determining Rank of Matrix:

(1) Find the rank of the matrix A - $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 5 \end{bmatrix}$.

(2) Find the rank of the matrix A = $\begin{bmatrix} 6 & 2 & 0 & 4 \\ -2 & -1 & 3 & 4 \\ -1 & -1 & 6 & 10 \end{bmatrix}$

(3) Determine the rank of the following matrix: B = $\begin{bmatrix} 2 & -1 & 3 & 4 \\ 0 & 3 & 4 & 1 \\ 2 & 3 & 7 & 5 \\ 2 & 5 & 11 & 6 \end{bmatrix}$,

C = $\begin{bmatrix} 0 & 1 & -3 & -1 \\ 1 & 0 & 1 & 1 \\ 3 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 \end{bmatrix}$

Determining Linearly dependent/independent: Given in Lecture note + Exercises – (Page – 271): 1, 2, 3, 6, 15